

List of questions for the midterm

1. Inverse function theorem.
2. Sard's lemma.
3. Degree modulo 2 and integer degree; their homotopy invariance.
4. Construction of partition of unity.
5. Brouwer fixed-point theorem.
6. Thom's transversality theorem, intersection number.
7. Whitney embedding theorem (for closed manifolds).
8. Vector fields as sections of tangent bundle: integral curves, flows, straightening lemma.
9. Vector fields as a differential operator: Lie bracket, Jacobi identity.
10. Straightening lemma for commuting vector fields.
11. Lie derivative of tensor fields: definitions and proof of identities

$$\begin{aligned}\mathcal{L}_X(\alpha \otimes \beta) &= (\mathcal{L}_X\alpha) \otimes \beta + \alpha \otimes (\mathcal{L}_X\beta), \\ \mathcal{L}_X \text{Contraction} &= \text{Contraction} \mathcal{L}_X, \\ \mathcal{L}_{X+Y} &= \mathcal{L}_X + \mathcal{L}_Y, \\ \mathcal{L}_X \mathcal{L}_Y - \mathcal{L}_Y \mathcal{L}_X &= \mathcal{L}_{[X,Y]}.\end{aligned}$$

12. Grassmann algebra and its existence (tensor interpretation).
13. Differential forms: definition, Lie, external, and internal derivative, their product rules, pullback and its relation to wedge product and external differential ($f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta$ and $f^*d = df^*$).
14. Cartan's magic formula.
15. Stokes' theorem, closed and exact forms.
16. De Rham cohomology algebra: definitions, an example of calculations via symmetry.

- 17. Homotopy invariance of De Rham cohomology, Poincaré's lemma.
- 18. Mayer–Vietoris sequence: formulation + an application.
- 19. Top cohomology. Cohomological definition of degree.
- 20. Moser's theorem via Moser's trick.
- 21. Morse theory: degenerate and nondegenerate critical points, existence of Morse function (for closed manifolds).
- 22. Morse theory: product structure at noncritical levels.
- 23. Morse lemma.
- 24. Handle decomposition: rearrangement of handles (reordering critical levels according to the index), handle body decomposition of 3-manifolds (Heegaard splitting).